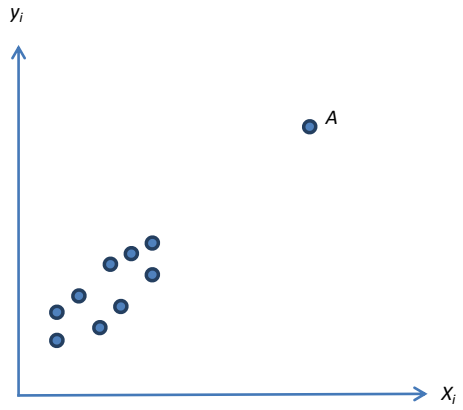
Chapter 5

Diagnostic for Leverage and Influence

The location of observations in x-space can play an important role in determining the regression coefficients. Consider a situation like in the following figure:

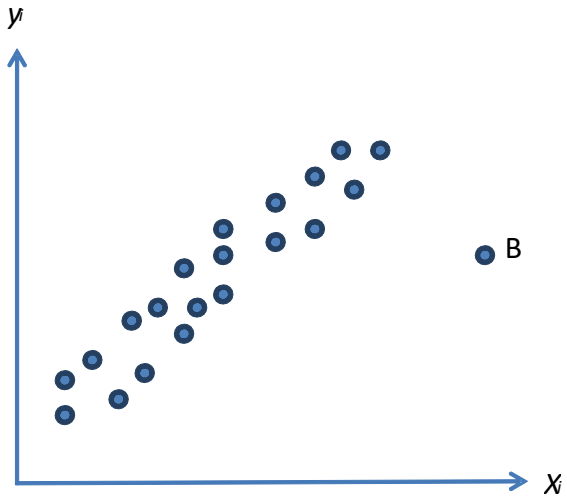


The point A in this figure is remote in $x-$space from the rest of the sample, but it lies almost on the regression line passing through the rest of the sample points. This is a **leverage point.**

It is an unusual x-value and may control certain model properties.

- This point does not affect the estimates of the regression coefficients.
- It affects the model summary statistics, e.g., R^2, standard errors of regression coefficients, etc.

Now consider point B in the following figure:



This point has a moderately unusual x-coordinate and the y-value is also unusual. This is an **influential point**

- It has a noticeable impact on the model coefficients.
- It pulls the regression model in its direction.

Sometimes, a small subset of data can exert disproportionate influence on the model's coefficients and properties. In an extreme case, the parameter Estimates may depend more on the influential subset of points than on the majority of the data. This is an undesirable situation. A regression model must represent all the sample observations, not just a few. We would like to identify these influential points and assess their impact on the model.

- If these influential points are "bad" values, they should be eliminated from the sample.
- If nothing is wrong with these points, but if they control the model properties, then it is necessary to determine how they affect the regression model in use.

Leverage

The location of points in x-space affects the model properties like parameter estimates, standard errors, predicted values, summary statistics, etc. The hat matrix $H = X(X'X)^{-1}X'$ plays an important role in identifying influential observations. Since

$$V(\hat{y}) = \sigma^2 H$$
$$V(e) = \sigma^2 (I - H),$$

($\hat{y}$ is the fitted value and e is residual) the elements h_{ii} of H may be interpreted as the amount of leverage exerted by the ith observation y_i on the ith fitted value $\hat{y}_i$.

The ith diagonal element of H is

$$h_{ii} = x_i'(X'X)^{-1}x_i$$

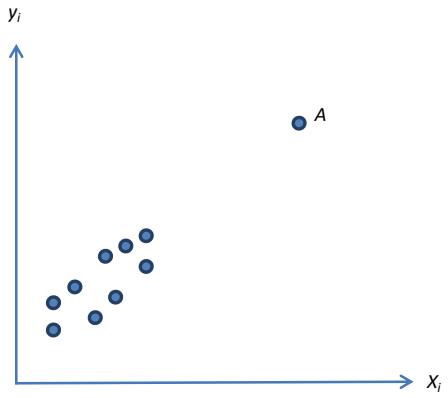
where x_i' is the ith row of X-matrix. The hat matrix diagonal is a standardized measure of the distance of ith observation from the centre (or centroid) of the $x-$space. Thus, large hat diagonals reveal observations that are potentially influential because they are remote in x-space from the rest of the sample.

The average size of the hat diagonal $(\bar{h})$ is

$$\bar{h} = \frac{\sum h_{ii}}{n}$$
$$= \frac{trH}{n}$$
$$= \frac{k}{n}$$

- If $h_{ii} > 2\bar{h} = \frac{2k}{n} \Rightarrow$ the point is remote enough from the rest of the data to be considered as a leverage point.
- Care is needed in using the cutoff value $\frac{2k}{n}$ and the magnitudes of k and n are to be assessed. There can be situations where $\frac{2k}{n} > 1$ and then this cut-off does not apply.

Not all leverage points are influential on the regression coefficients. In the following figure



the point A

- will have a large hat diagonal and is surely a leverage point.
- has no effect on the regression coefficients, as it lies on the same line that passes through the remaining observations.

The hat diagonal examines only the location of observations in x-space, so we can look at the studentized residual or R-student in conjunction with the h_{ii}.

Observation with

- large hat diagonal and
- large residuals

are likely to be influential.

Measures of influence

(1) Cook's D-statistics

If the data set is small, then the deletion of values greatly affects the fit and statistical conclusions.

In measuring influence, it is desirable to consider both

- the location of the point is x-space and
- the response variable.

The Cook's distance statistics, denoted as Cook's D-statistic, is a measure of the distance between the least-squares estimate based on all n observations in b and the estimate obtained by deleting the ith point, say $b_{(i)}$. It is given by

$$D_i(M,C) = \frac{(b_{(i)} - b)'M(b_{(i)} - b)}{C}; \quad i = 1,2,...,n.$$

The usual choice of M and C are

$$M = X'X$$
$$C = kMS_{res}$$

So

$$D_i(X'X, kMS_{res}) = \frac{(b_{(i)} - b)'X'X(b_{(i)} - b)}{kMS_{res}}; \quad i = 1,2,...,n$$
$$= \frac{(\hat{y} - \hat{y}_{(i)})'(\hat{y} - \hat{y}_{(i)})}{kMS_{res}}$$

where

$$\hat{y} = Xb$$
$$\hat{y}_{(i)} = Xb_{(i)}$$
$$b = (X'X)^{-1}X'y.$$

Points with large $D_i \Rightarrow$ the points have a considerable influence on OLSE b.

Since

$$\frac{(b_{(i)} - b)'X'X(b_{(i)} - b)/k}{SS_{res}/(n-k)}$$

looks like a statistic having a $F(k, n-k)$ distribution. Note that these statistics do not have a $F(k, n-k)$ distribution.

So the magnitude of D_i is assessed by comparing it with $F_\alpha(k, n-k)$. If $D_i = F_{0.5}(k, n-k),$ then deleting the point i would move $\hat{\beta}_{(i)}$ to the boundary of an approximate 50% confidence region for β based on the complete data set.

This displacement is large and indicates that the OLSE is sensitive to the ith data point.

- Since $F_{0.5}(k, n-k) \approx 1,$ we usually consider that points for which $D_i > 1$ to be influential.
- Ideally, each $b_{(i)}$ is expected to stay within the boundary of a 10-20% confidence region.
- D_i is not an F-statistic, but cut off at 1 works very well in practice.

Since

$$D_i = \frac{(\hat{y}_{(i)} - \hat{y})'(\hat{y}_{(i)} - \hat{y})}{kMS_{res}},$$

so D_i can be interpreted as a squared Euclidean distance (apart from kMS_{res}) that the vector of fitted values moves when the ith observation is deleted.

Since

$$b_i - b_{(i)} = \frac{(X'X)^{-1}x_i e_i}{1 - h_{ii}},$$

the D_i can be written as

$$\begin{aligned} D_i &= \frac{(b_i - b_{(i)})'X'X(b_i - b_{(i)})}{kMS_{res}} \\ &= \frac{x_i'(X'X)^{-1}(X'X)(X'X)^{-1}x_i e_i^2}{(1-h_{ii})^2 kMS_{res}} \\ &= \left(\frac{e_i}{1-h_{ii}}\right)^2 \left(\frac{h_{ii}}{kMS_{res}}\right) \\ &= \frac{r_i^2}{k}\left(\frac{h_{ii}}{1-h_{ii}}\right) \end{aligned}$$

where r_i is studentized residual.

D_i : product of squared i^{th} studentized residual and $h_{ii}/(1-h_{ii})$.

: Reflects how well the model fits the ith observation y_i and a component that measures how far that point is from the rest of the data.

: Either component or both may contribute to a large value of D_i.

: Thus D_i combines residual magnitude for i^{th} observation and location of points in x - space to assess influence.

(2) *DFBETAS* and *DFFITS*

Cook's distance measure is a deletion diagnostic, i.e., it measures the influence of ith observation if it is removed from the sample.

There are two more statistics:

(I) *DFBETAS*

DFBETAS indicates how much the regression coefficient changes if the ith observation were deleted. Such a change is measured in terms of standard deviation units. This statistic is

$$DFBETAS_{j,i} = \frac{b_j - b_{j(i)}}{\sqrt{S^2_{(i)} C_{jj}}}$$

where C_{jj} is the jth diagonal element of $(X'X)^{-1}$ and $b_{j(i)}$ is the jth regression coefficient computed without the use of ith observation.

Large (in magnitude) value of $DFBETAS_{j,i}$, indicates that i^{th} observation has considerable influence on the jth regression coefficient. The values of $DFBETAS_{j,i}$ can be expressed in a $n \times k$ matrix that conveys similar information to the composite influence information in Cook's distance measure.

The computation of $DFBETAS_{j,i}$ is as follows:

Define the $k \times n$ matrix $R = (X'X)^{-1}X'$. The n elements in the jth row of R produce the leverage that the n observations in the sample have on $\hat{\beta}_j$.

$DFBETAS_{j,i}$ is the jth element of $(b - b_{(i)})$ divided by a standardization factor

$$b_i - b_{(i)} = \frac{(X'X)^{-1} x_i' e_i}{1 - h_{ii}}.$$

The jth element of $(b_i - b_{(i)})$ can be expressed as

$$b_i^j - b_{(i)}^j = \frac{r_{j,i} e_i}{1 - h_{ii}}.$$

Let $r_{j,i} = ((R))$ denotes the elements of R, so

$$\begin{aligned}(RR')' &= \left[(X'X)^{-1}X'X(X'X)^{-1}\right]' \\ &= (X'X)^{-1} \\ &= C \\ &= R'R.\end{aligned}$$

Since

$$C_{jj} = r_j' r_j,$$

so

$$\sqrt{S^2_{(i)}C_{jj}} = \sqrt{S^2_{(i)}r_j'r_j}$$

$$DFBETAS_{j,i} = \frac{b^j_i - b^j_{(i)}}{\sqrt{S^2_{(i)}C_{jj}}}$$

$$= \left(\frac{r_{j,i}e_i}{1-h_{ii}}\right)\frac{1}{\sqrt{S^2_{(i)}r_j'r_j}}$$

$$= \frac{r_{j,i}}{\sqrt{r_j'r_j}}\frac{t_i}{\sqrt{1-h_{ii}}}$$

$$\downarrow \qquad \downarrow$$

Measures leverage (impact of ith observation on b_i	Measures effect of large residuals

where t_i is the ith R-student residual. Now if $\left|DFBETAS_{j,i}\right| > \frac{2}{\sqrt{n}}$, then i^{th} observation warrants examination.

(II) *DFFITS*

The deletion influence of ith observation on the predicted or fitted value can be investigated by using the diagnostic by Belsley, Kuh and Welsch as

$$DFFITS_i = \frac{\hat{y}_i - \hat{y}_{(i)}}{\sqrt{S^2_{(i)}h_{ii}}},\ i = 1,2,...,n$$

where $\hat{y}_{(i)}$ is the fitted value of y_i obtained without the use of the ith observation. The denominator is just a standardization, since $Var(\hat{y}_i) = \sigma^2 h_{ii}$.

This $DFFITS_i$ is the number of standard deviations that the fitted value $\hat{y}_i$ changes of ith observation is removed.

Computationally,

$$DFFITS_{\text{i}} = \sqrt{\frac{h_{ii}}{1-h_{ii}}}\frac{e_i}{S_{(i)}\sqrt{1-h_{ii}}}$$
$$= t_i\sqrt{\frac{h_{ii}}{1-h_{ii}}}$$
$$= R-\text{student} \times \text{leverage of } i^{th} \text{ observation}$$

where t_i is R-student.

- If the data point is an outlier, then R-Student will be large in magnitude.
- If the data point has high leverage, then h_{ii} will be close to unity.
- In either of these cases, $DFFITS_i$ can be large.
- If $h_{ii} \approx 0,$ then the effect of R-Student will be moderated.
- If R-student is near to zero, then combined with a high leverage point, then $DFFITS_i$ can be a small value.
- Thus $DFFITS_i$ is affected by both leverage and prediction error. Belsley, Kuh and Welsch suggest that any observation for which

  $$\left|DFFITS_{\text{i}}\right| > 2\sqrt{\frac{k}{n}}$$

  warrants attention.

Note: The cutoff values of $DFFITS_{j,i}$ and $DFFITS_i$ are only guidelines. It is very difficult to provide cutoffs that are correct for all cases. So an analyst is recommended to utilise information about both the diagnostic methods and the application environment when selecting a cutoff.

For example, if $DFFITS_i = 1,$ say, we could translate this into actual response units to determine just how much $\hat{y}_i$ is affected by removing the ith observation.

Then use $DFFITS_{j,i}$ to see whether this observation is responsible for the significance (or perhaps nonsignificance) of particular coefficients or for changes in the sign of a regression coefficient.

$DFFITS_{j,i}$ can be used to determine how much change in actual problem-specific units a data point has on the regression coefficient. Sometimes these changes will be of importance in a problem-specific context, even though the diagnostic statistics do not exceed the formal cutoff.

The recommended cutoffs are a function of sample size n. Indeed, any formal cutoff should be a function of sample size. However, in practice, these cutoffs often identify more data points than an analyst may wish to analyze. This is particularly true in small samples. The cutoff values provided by Belsley, Kuh and Welsch make more sense for large samples. When n is small, then diagnostic views are preferred.

A measure of model performance generalized variance

The diagnostics D_i, $DFFITS_{j,i}$ and $DFFITS_i$ provide insight into the effect of observations on the estimated coefficient $\hat{\beta}_j$ and fitted values $\hat{y}_i$. They do not provide any information about the overall precision of estimation.

The **generalized variance** is defined as the determinant of the covariance matrix and is a convenient scalar measure of precision. The generalized variance of OLSE b is

$$GV(b) = |V(b)| = \left|\sigma^2 (X'X)^{-1}\right|.$$

To express the role of ith observation on the precision of estimation, define

$$COVRATIO_i = \frac{\left|(X'_{(i)}X_{(i)})^{-1} S^2_{(i)}\right|}{\left|(X'X)^{-1} MS_{res}\right|}, \quad i = 1, 2, ..., n.$$

If $COVRATIO_i > 1 \Rightarrow i^{th}$ observation improves the precision of estimation.

If $COVRATIO_i < 1 \Rightarrow$ the inclusion of ith observation degrades the precision computationally,

$$COVRATIO_i = \frac{\left(S^2_{(i)}\right)^k}{MS^k_{res}} \left(\frac{1}{1-h_{ii}}\right) \quad \text{where} \quad \frac{1}{1-h_{ii}} = \frac{\left|(X'_{(i)}X_{(i)})^{-1}\right|}{\left|(X'X)^{-1}\right|}.$$

- So a high leverage point will make $COVRATIO_i$ large. This is logical since a high-leverage point will improve the precision, unless the point is an outlier in y-space.
- If i^{th} observation is an outlier, then $\frac{\left(S^2_{(i)}\right)^k}{MS^k_{res}}$ will be much less than unity.

- Cutoff values for *COVRATIO* are not easy to obtain. It is suggested that

  if $COVRATIO_i > 1 + \frac{3k}{n}$

  or if $COVRATIO_i < 1 - \frac{3k}{n}$,

  then i^{th} point should be considered influential. The lower bound is only appropriate when $n > 3k$.

These cutoffs are only recommended for large samples.